

MCP Guardian ? Technical Overview

Version: 3.4.1

Package

Confidentiality: This document describes product architecture at a high level. It omits signing keys, internal threat research heuristics, and customer-specific configurations.

1. Executive summary

MCP Guardian is a governance proxy for the Model Context Protocol (MCP). It sits between AI clients (Cursor, Claude Desktop, Cline, Windsurf, and others) and upstream MCP tool servers. Every tools/call request is evaluated against customer-controlled policy before reaching real infrastructure (filesystem, shells, APIs, databases).

The product delivers three outcomes in one install:

1. Enforcement ? deterministic allow/block with explainable reasons

2. Observability

All capabilities ship in the open-source / npm distribution; there is no separate paid tier for core security features.

2. Problem and placement

Without Guardian | With Guardian

IDE connection

Guardian is the control plane for agent tool access?analogous to an API gateway for LLM tool calls, optimized for MCP JSON-RPC semantics.

3. System architecture

3.1 Runtime topology (typical)

A standard developer deployment runs one Node.js process (?18) that hosts:

Layer | Responsibility

Proxy

Upstream: child-process MCP servers (stdio) or HTTP/SSE relays to remote MCP hosts.

Clients: AI assistants configured via mcp-guardian onboard to use Guardian as the MCP transport.

3.2 Tool-call governance pipeline

Blocked requests never reach upstream. The hot path is:

1. Transport receives tools/call JSON-RPC
2. Pre-for

Rate-limit counters persist across policy hot-reload via an in-process shared store (Redis in multi-replica enterprise mode).

3.3 Offline and learning pipelines

These run beside the live proxy (not in the per-request hot path):

Pipeline | Purpose

**Adversa

Outputs feed policy refinement and harness corpus growth; no silent production policy changes.

4. Main components

Component | Location (repo) | Description

CLI | `S

5. Key technologies

Category | Technology

**Languag

6. Deployment models

6.1 Developer / single-user (default)

`npm install -g @mcp-guardian/server`

Setting | Default

6.2 Team / CI

? Run pnpm harness for offline policy regression

6.3 Enterprise / multi-replica

Documented in docs/ENTERPRISE_DEPLOYMENT.md and Helm charts under deploy/helm/:

Requirement | Notes

Kubernetes Helm values: deploy/helm/mcp-guardian/. Multi-region notes: docs/MULTI_REGION.md.

6.4 Cloud marketing / docs site

Static site and docs hosted on Vercel (mcp-guardian-cloud.vercel.app). Production MCP governance runs on customer infrastructure (local proxy or customer K8s)?not as a mandatory SaaS middleman for tool traffic.

7. Security and privacy posture (summary)

This document does not enumerate specific detection signatures or internal scoring weights.

8. Integration and extension points

Extension | Mechanism

Custom p

9. Evaluation flow (recommended)

For technical or investor evaluation, focus on the wrap ? govern ? observe loop:

1. Onboard ? mcp-guardian onboard rewrites IDE configs to Guardian-wrapped servers

2. Traffic

Primary demo URL: http://localhost:4000 when running pnpm dashboard:proxy.

10. References

Resource | URL / path

README

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